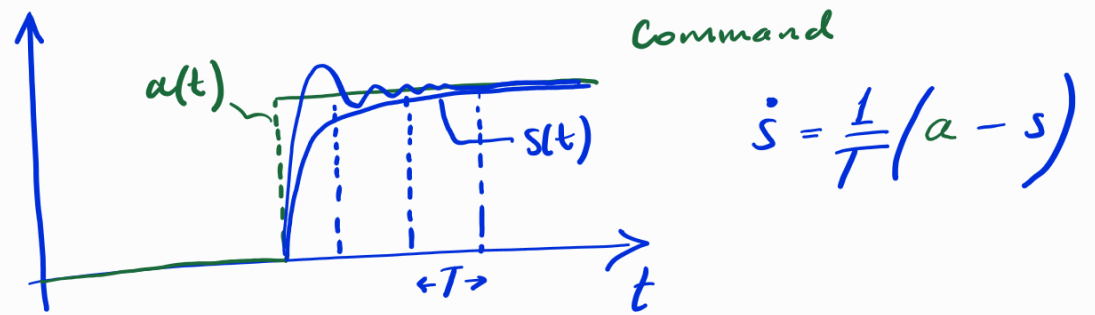


Essentials of backstepping



Pendulum plant

$$\dot{s}_1 = s_2$$

$$\dot{s}_2 = -\frac{g}{l} \sin s_1 + \frac{a}{ml^2}$$

τ

Motor dynamics: $\dot{\tau} = \frac{1}{T}(-\tau + a)$ new command

$$s_3 := \tau$$

Plant becomes

$$\dot{s}_1 = s_2$$

$$\dot{s}_2 = -\frac{g}{l} \sin s_1 + \frac{s_3}{ml^2}$$

$$\dot{s}_3 = \frac{1}{T}(-\tau + a)$$

We know how to control the base plant, say, by a policy π_0 .

Suppose, we designed a certifying certificate

L_0 s.t.

$$\dot{L}_0(s | \pi_0) \leq -\kappa_0(\|s\|)$$

Backstepping idea:

$$L_1 := L_0 + \frac{1}{2} \| s_3 - \pi_0(s_1, s_2) \|^2$$

Simplest structure

$$\dot{s} = f(s) + g(s)\phi$$

$$\dot{\phi} = a'$$

We have now:

$$\dot{s}_1 = s_2$$

$$\dot{s}_2 = -\frac{g}{l} \sin s_1 + \frac{s_3}{m l^2}$$

$$\dot{s}_3 = \frac{1}{T} (-s_3 + a)$$

We can do a coord. transf-n

$$\phi := s_3$$

$$a' := \frac{1}{T} (-\phi + a)$$

More general case:

$$\dot{s} = f(s) + g(s)\phi$$

$$\dot{\phi} = f_1(s, \phi) + g_1(s, \phi)a$$

Let's assume the following structure:

$$\dot{s} = f(s) + g(s)\phi$$

$$\dot{\phi} = a$$

(π_0, L_0, κ_0) was a stabilization triple for the base plant

$$L_1 := L_0 + \frac{1}{2} \|\phi - \pi_0(s)\|^2$$

$$\dot{L}_1 = \underline{L_f L_0} + \underline{L_g L_0 \phi} + (\phi - \pi_0(s))^T (\underline{a} - L_f \pi_0 - L_g \pi_0 \phi)$$

if π_0 matched ϕ here, we'd have recovered $-K_0(\|s\|)$

Suggestion for the new action:

$$a \leftarrow \pi_1(s, \phi) := - \underset{\substack{\uparrow \\ \text{control gain}}}{K} (\phi - \pi_0(s))$$

Then,

$$\dot{L}_1 = L_f L_0 + L_g L_0 \phi - K \|\phi - \pi_0(s)\|^2 - (\phi - \pi_0(s))^T (L_f \pi_0 + L_g \pi_0 \phi)$$

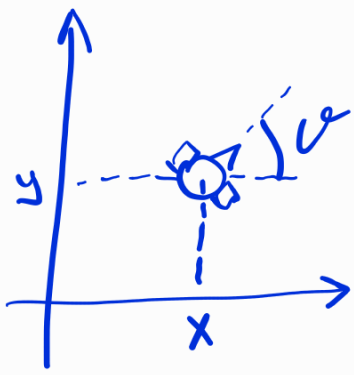
To get a valid decay rate, we should augment the policy as follows:

$$a \leftarrow \pi_1'(s, \phi) := -K(\phi - \pi_0(s)) - L_g L_0 + L_{f+g\phi} \pi_0$$

$$\text{Hence, } \dot{L}_1 \leq -K_0(\|s\|) - K \|\phi - \pi_0(s)\|^2$$

Verify that the following function is a CLF for the mobile robot:

$$L = \frac{1}{2} \left(x^2 + y^2 + \left(\psi - \arctan \frac{y}{x} \right)^2 \right)$$



$$\begin{aligned}\dot{x} &= r \cos \theta \\ \dot{y} &= r \sin \theta \\ \dot{\theta} &= \omega\end{aligned}$$

$$a = \begin{pmatrix} \dot{r} \\ \omega \end{pmatrix}$$

$$s = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix}$$

CLF: $\forall s \exists a \quad L_{P(s,a)} L < 0$

for a plant $\dot{s} = P(s, a)$

Extended plant:

$$\begin{aligned}\dot{x} &= r \cos \theta \\ \dot{y} &= r \sin \theta \\ \dot{\theta} &= \omega \\ \dot{r} &= \frac{1}{m} F \\ \dot{\omega} &= \frac{1}{J} \tau\end{aligned}$$

$$s' = \begin{pmatrix} x \\ y \\ \theta \\ r \\ \omega \end{pmatrix}, \quad a' = \begin{pmatrix} F \\ \tau \end{pmatrix}$$